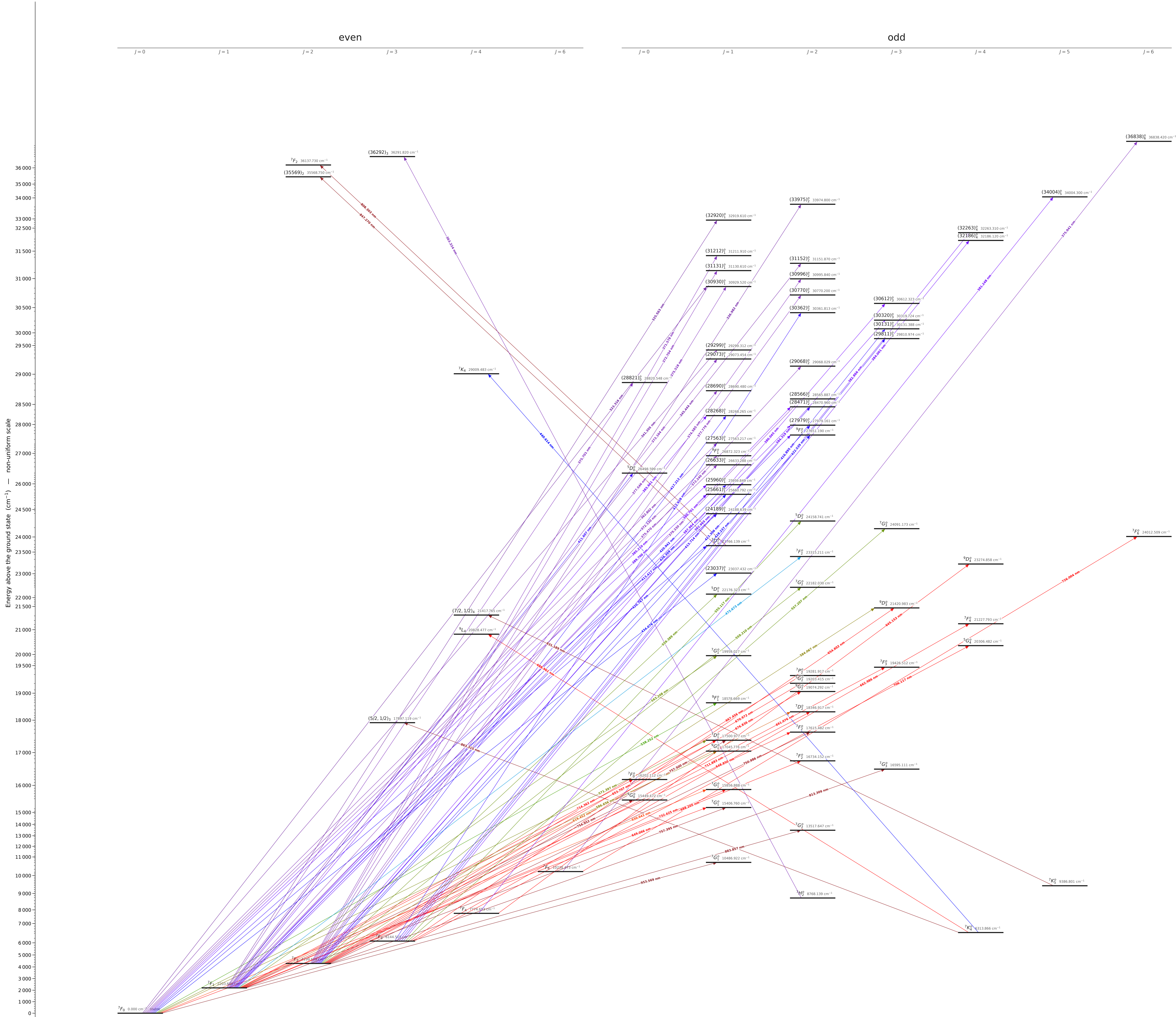
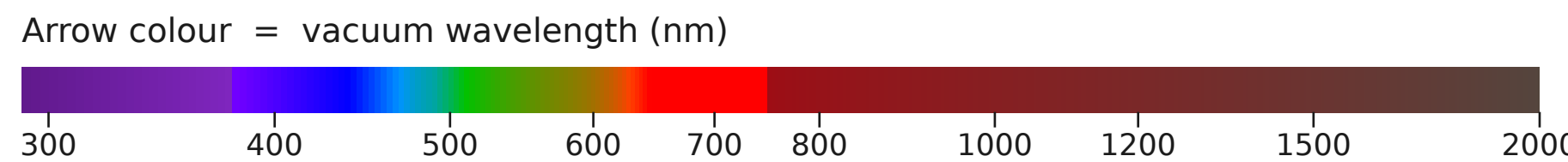


Plutonium

Pu (Pu I). Classified transitions below 2 μm (levels and lines from the literature): 80 levels, 85 lines.

Pu⁺ ionisation limit: 48601.00 cm⁻¹ = 6.02576 eV (205.757 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle J || e r || J' \rangle$ in e a₀
Matrix elements are measured or NIST values: * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (from the cited literature: NIST lists no excited levels) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\langle U e r U' \rangle$ (e a ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
⁷ F ₀ - (² 5960) ^o	385.2103	385.1010	778.2567	-	-	-	-	385.210 nm line used for high-resolution in-source resona...
⁷ F ₁ - (² 7979) ^o	387.9645	387.8546	772.7317	-	-	-	-	387.965 nm line from the thermally populated 7F ₁ level, u...
⁷ F ₀ - (² 0 ^o)	413.4172	413.3006	725.1572	-	-	-	-	first step of the Mainz two-step resonance-ionization sch...
⁷ F ₀ - (² 3037) ^o	420.7667	420.6482	712.4909	-	-	-	-	first step
⁷ F ₀ - (² 3037) ^o	434.0762	433.9542	690.6448	-	-	-	-	alternative uranium-free first step
⁷ D ^o ₁ - ⁷ F ₂	808.3035	808.0813	370.8910	-	-	-	-	second step
⁷ D ^o ₁ - (³ 5569) ₁	847.2702	847.0374	353.8334	-	-	-	-	second step

λ, ν from the level energies of the cited literature (NIST lists no excited levels) unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2τ₁).

²⁴¹Pu hyperfine structure (I = 5/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
⁷ F ₀ - (² 5960) ^o	-2(5)	1059(8)	F=3/2: +748.72 F=5/2: -844.68 F=7/2: +259.15
⁷ F ₁ - (² 7979) ^o	-278(6)	101(18)	F=3/2: +1043.91 F=5/2: +197.46 F=7/2: -670.05
⁷ F ₂ - (² 9096) ^o	38(3)	-175(14)	F=1/2: -386.68 F=3/2: -251.20 F=5/2: -69.25 F=7/2: +93.39 F=9/2: +144.65

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
⁷ F ₀ - (³ 0930) ^o	323.316	240-239	-2218.00
⁷ F ₂ - (³ 3975) ^o	336.982	240-239	-2398.00
⁷ F ₀ - (² 9299) ^o	341.305	240-239	-1649.00
⁷ F ₁ - (³ 1152) ^o	345.444	240-239	-3538.00
⁷ F ₂ - (² 7563) ^o	362.802	240-239	-2218.00
⁷ H ^o ₂ - (³ 6292) ₁	363.324	240-239	-7525.00
⁷ F ₂ - (³ 1212) ^o	371.578	240-239	-2308.00
⁷ F ₀ - ⁹ F ^o ₁	372.130	240-239	-2069.00
⁷ F ₁ - (² 9073) ^o	372.164	240-239	-1139.00
⁷ F ₁ - (² 9068) ^o	372.240	240-239	-1289.00
⁷ F ₂ - (³ 1131) ^o	372.704	240-239	-2188.00
⁷ F ₂ - (³ 0996) ^o	374.585	240-239	-2428.00
⁷ F ₀ - (² 6633) ^o	375.470	240-239	-3657.00
⁷ F ₂ - (³ 0930) ^o	375.518	240-239	-2368.00
⁷ F ₀ - (³ 6838) ^o	375.941	240-239	-3208.00
⁷ F ₁ - (² 8690) ^o	377.546	240-239	-480.00
⁷ F ₂ - (³ 0770) ^o	377.779	240-239	-2159.00
⁷ F ₁ - (² 8566) ^o	379.330	240-239	-3418.00
⁷ F ₂ - (³ 0612) ^o	380.045	240-239	-2698.00
⁷ F ₁ - (² 8471) ^o	380.701	240-239	-3448.00
⁷ F ₄ - (³ 4004) ^o	381.248	240-239	-2338.00
⁷ F ₂ - (³ 2263) ^o	382.866	240-239	-1739.00

Shift = ν(first isotope) - ν(second isotope).